Michael Diaz-Stewart

Vancouver, BC | Relocating to Toronto | diaz-@live.ca
Portfolio website: https://mikediaz1.github.io/ | GitHub | Kaggle | LinkedIn

SUMMARY

Applied ML researcher and software developer with 3+ years of deep-learning research, specializing in medical computer vision, cross-institution evaluation, and reproducible PyTorch workflows. Prior commercial experience building web and mobile applications and deploying backend services on AWS.

EXPERIENCE

Research Assistant | AI in Medicine Lab, UBC — Jul 2023 – Oct 2026

- Built PyTorch pipelines for colorectal-cancer biomarker prediction from whole-slide images, combining four pretrained pathology foundation models with multiple-instance learning, patient-level aggregation, and ensembling.
- Evaluated generalization across TCGA and two external cohorts, investigating distribution shift, molecular-subtype label disagreement, and decision-threshold trade-offs.
- Evaluated a microsatellite instability (MSI) rule-out strategy at 94% sensitivity and 98% negative predictive value in retrospective external validation.
- Created reproducible, containerized GPU training and evaluation workflows using Singularity and SLURM.

Clinical ML Research Intern | St. Paul's Hospital — Jan 2024 – Aug 2024

- Proposed an AI screening framework for pacemaker lead extractions using clinical CT.
- Coordinated work with cardiac surgeons and presented at the UBC SBME Symposium.

Software Developer | Stafits — Nov 2017 – Oct 2018

- Built a React website and customer-management backend using Node.js, Express, and MongoDB; deployed backend services on AWS.
- Developed a React Native mobile application connected to the same backend and customer data as the web product.

PROJECTS

Self-supervised learning for multiplayer game data | Independent ML project

- Developed self-supervised models for League of Legends match data, comparing MLP and transformer encoders across masking strategies.
- Evaluated learned representations using held-out champion prediction and early-game performance tasks, with linear and MLP probes.

EDUCATION

Master of Applied Science, Biomedical Engineering | University of British Columbia — Thesis Defended Oct 2026

Grade: 93.2% | NSERC MUSIC Scholarship ($18,000); CANTRAIN Master's Studentship ($17,500)

Honours Bachelor of Science, Biology | York University — May 2022

First Class Standing; Dean's Honour Roll

PUBLICATIONS

1. Hadjifaradji A, **Diaz-Stewart M**, Chu J, et al. A Deep Learning Framework for Classification of Neuroendocrine Neoplasm Whole Slide Images. Cancers. 2025;17(18):2991. 10.3390/cancers17182991
2. Paul A, Mendis S, **Diaz-Stewart M**, et al. Comparison of Lanreotide and Octreotide LAR Use and Outcomes for Gastrointestinal Neuroendocrine Tumors in British Columbia, Canada. Cancer Control. 2026. 10.1177/10732748261417423